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 Studierichting: Informatica
 Oefeningenreeks 3G nr. 7.2

$$f(x) = -x^3 + 7x^2 - 8x$$

$$\text{dom } f : x \in \mathbb{R}$$

$$f(x) = x(-x^2 + 7x - 8)$$

$$x_1 = 0, x_2 = \frac{7 + \sqrt{17}}{2}, x_3 = \frac{7 - \sqrt{17}}{2}$$

$$f'(x) = -3x^2 + 14x - 8$$

$$x_1 = \frac{-14 + \sqrt{(-14)^2 - 4 \cdot (-3) \cdot (-8)}}{-6} = \frac{2}{3} \Rightarrow f\left(\frac{2}{3}\right) = \frac{-68}{27} \text{ (min)}$$

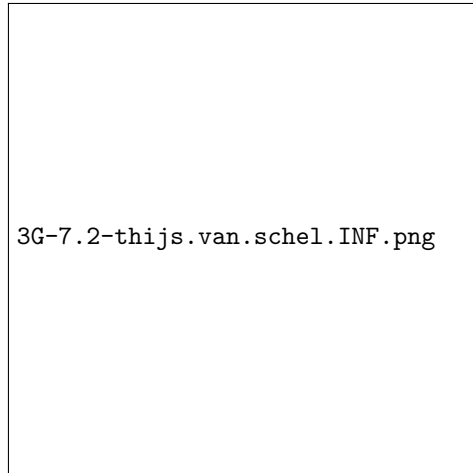
$$x_2 = \frac{-14 - \sqrt{(-14)^2 - 4 \cdot (-3) \cdot (-8)}}{-6} = 4 \Rightarrow f(4) = 16 \text{ (max)}$$

$$f''(x) = -6x + 14$$

$$x = \frac{7}{3} \Rightarrow f\left(\frac{7}{3}\right) = \frac{182}{27} \text{ (buigpunt)}$$

x	0			$\frac{2}{3}$	$\frac{7 - \sqrt{17}}{2}$			$\frac{7}{3}$	4			$\frac{7 + \sqrt{17}}{2}$	
$f(x)$	+	0	-	-	-	0	+	+	+	+	+	0	-
$f'(x)$	-	-	-	0	+	+	+	+	+	0	-	-	-
$f''(x)$	+	+	+	+	+	+	+	0	-	-	-	-	-
$f(x)$	$\searrow$	$\searrow$	$\searrow$	$m\left(\frac{-68}{27}\right)$	$\nearrow$	$\nearrow$	$\nearrow$	$\nearrow$	$\nearrow$	$M(16)$	$\searrow$	$\searrow$	$\searrow$
	$\frown$	$\frown$	$\frown$	$\frown$	$\frown$	$\frown$	$\frown$	$B\left(\frac{182}{27}\right)$	$\smile$	$\smile$	$\smile$	$\smile$	$\smile$

Tekening:



References